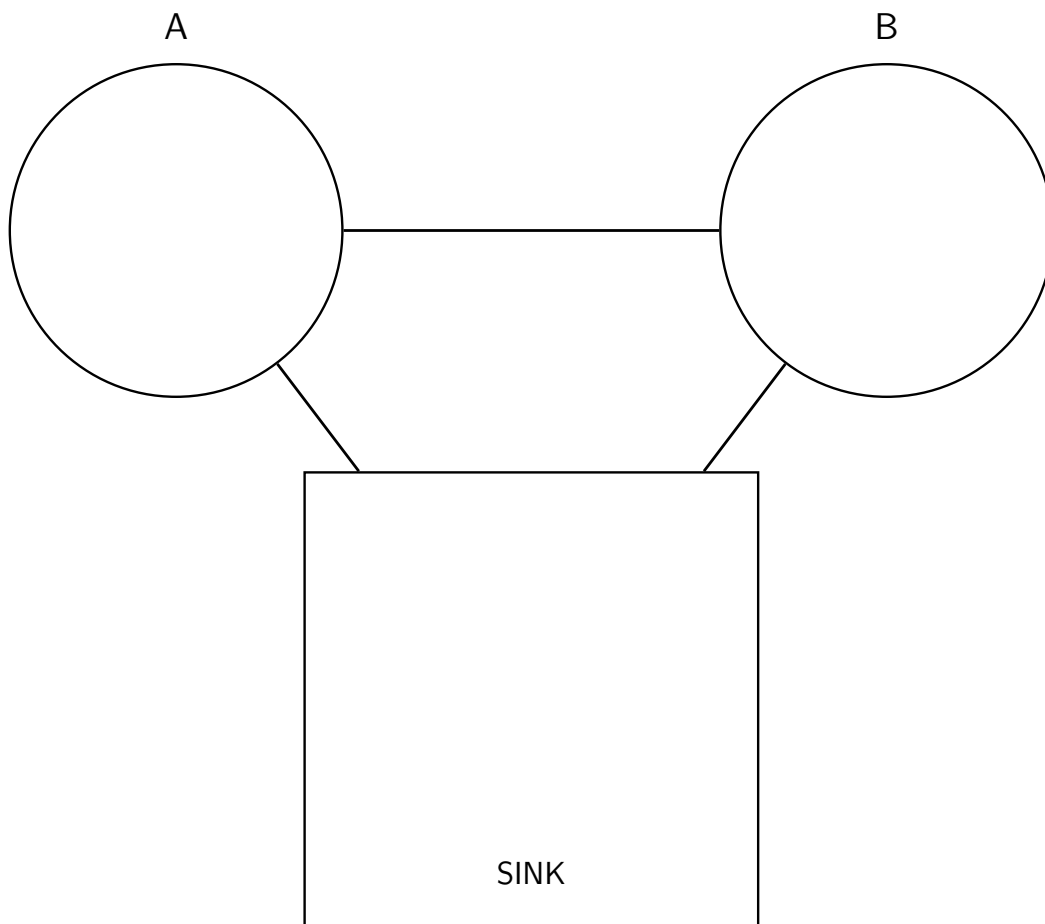


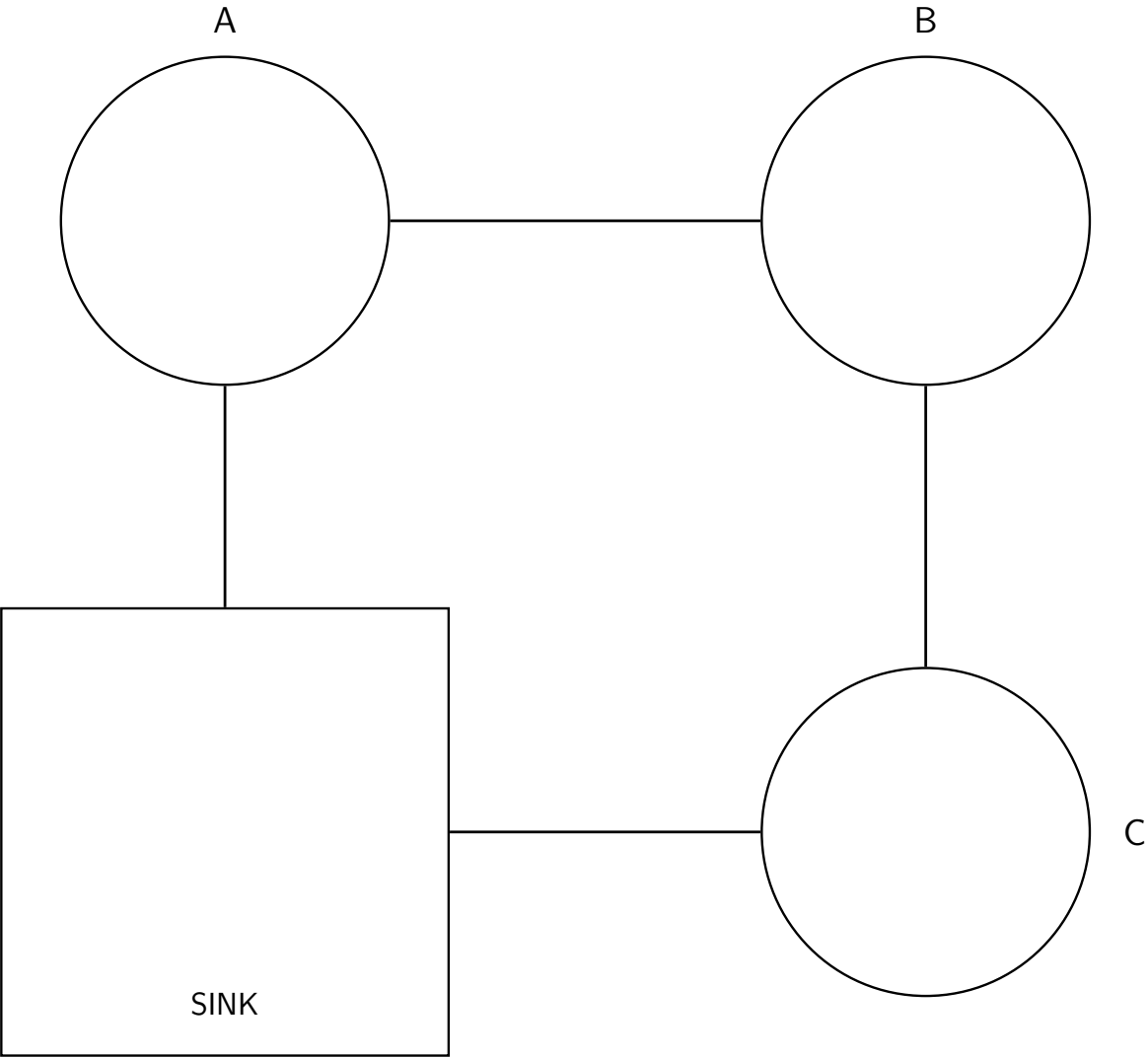
A circle needs one chip for every line touching it before it can share. To share, send one chip along each line. The sink keeps its chips and never shares. Choose one circle at a time. Stop when no circle can share. Empty the sink before each new start.

Problem 1: Finish each start in two different orders, if possible. Record the final piles and how many times each circle shares. Can a different order change either record?



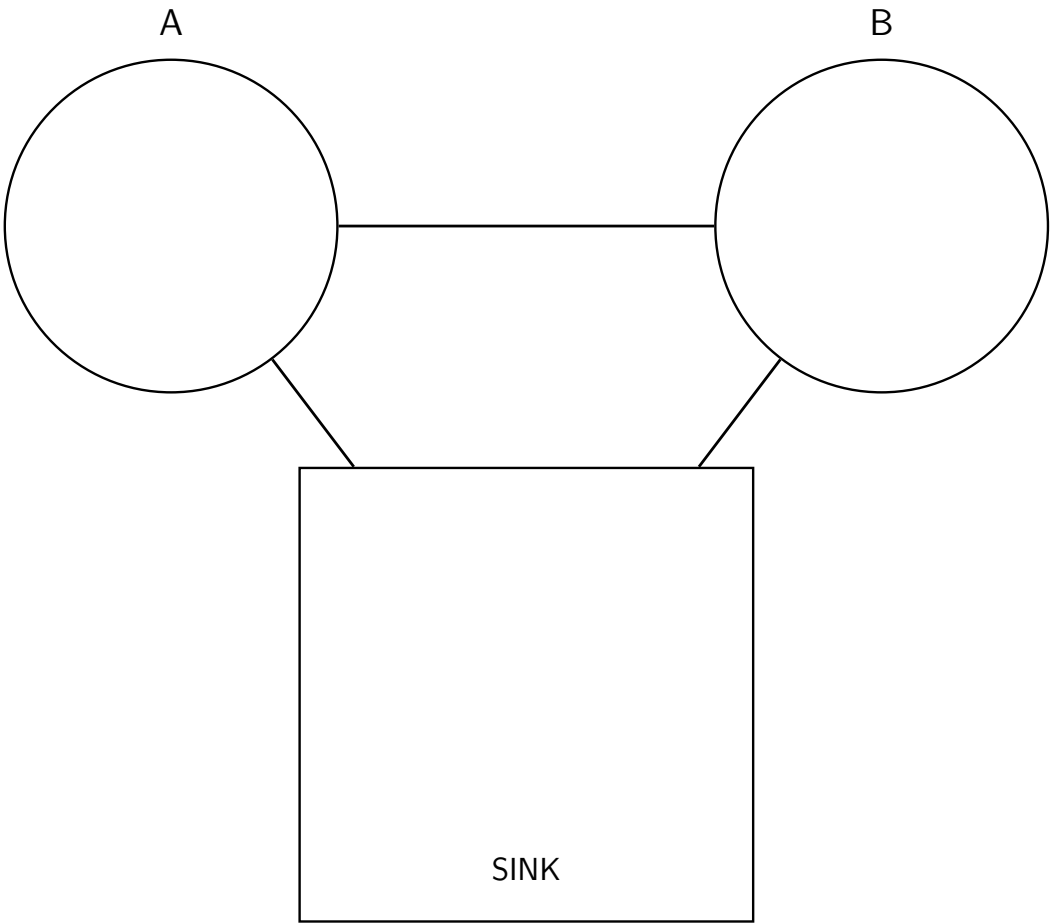
Start A	Start B	Finish A	Finish B	A shares	B shares
2	2				
2	2				
4	0				
4	0				
3	3				
3	3				

Problem 2: Finish each start in two different orders. Record the final piles and how many times each circle shares. Can either record change?



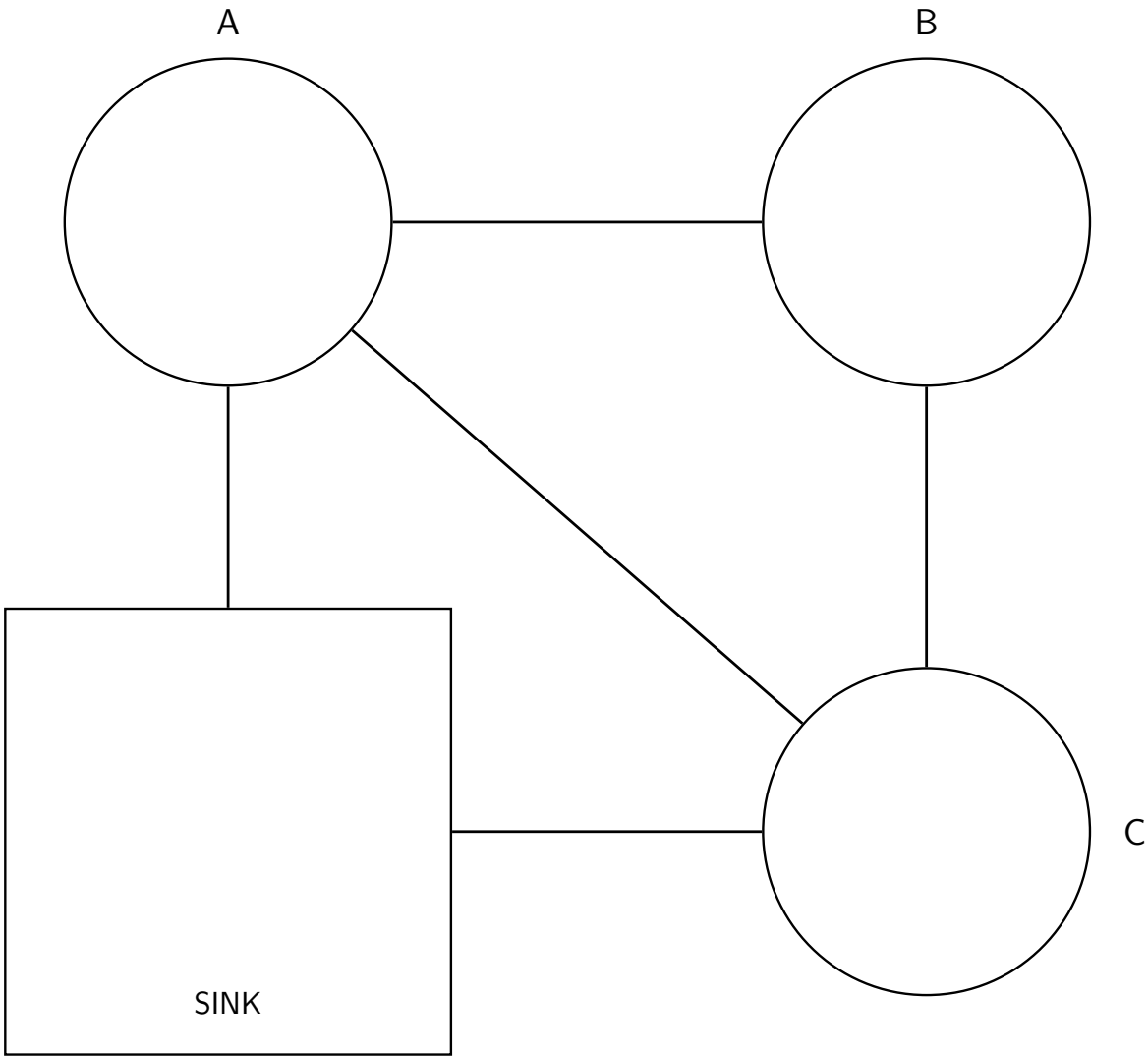
Start A	Start B	Start C	End A	End B	End C	A shares	B shares	C shares
0	4	0						
0	4	0						
2	1	2						
2	1	2						
2	4	2						
2	4	2						

Problem 3: Put exactly 6 chips altogether on A and B. Find every start that finishes with one chip at A and one at B. Explain how you know you have found every start that works.



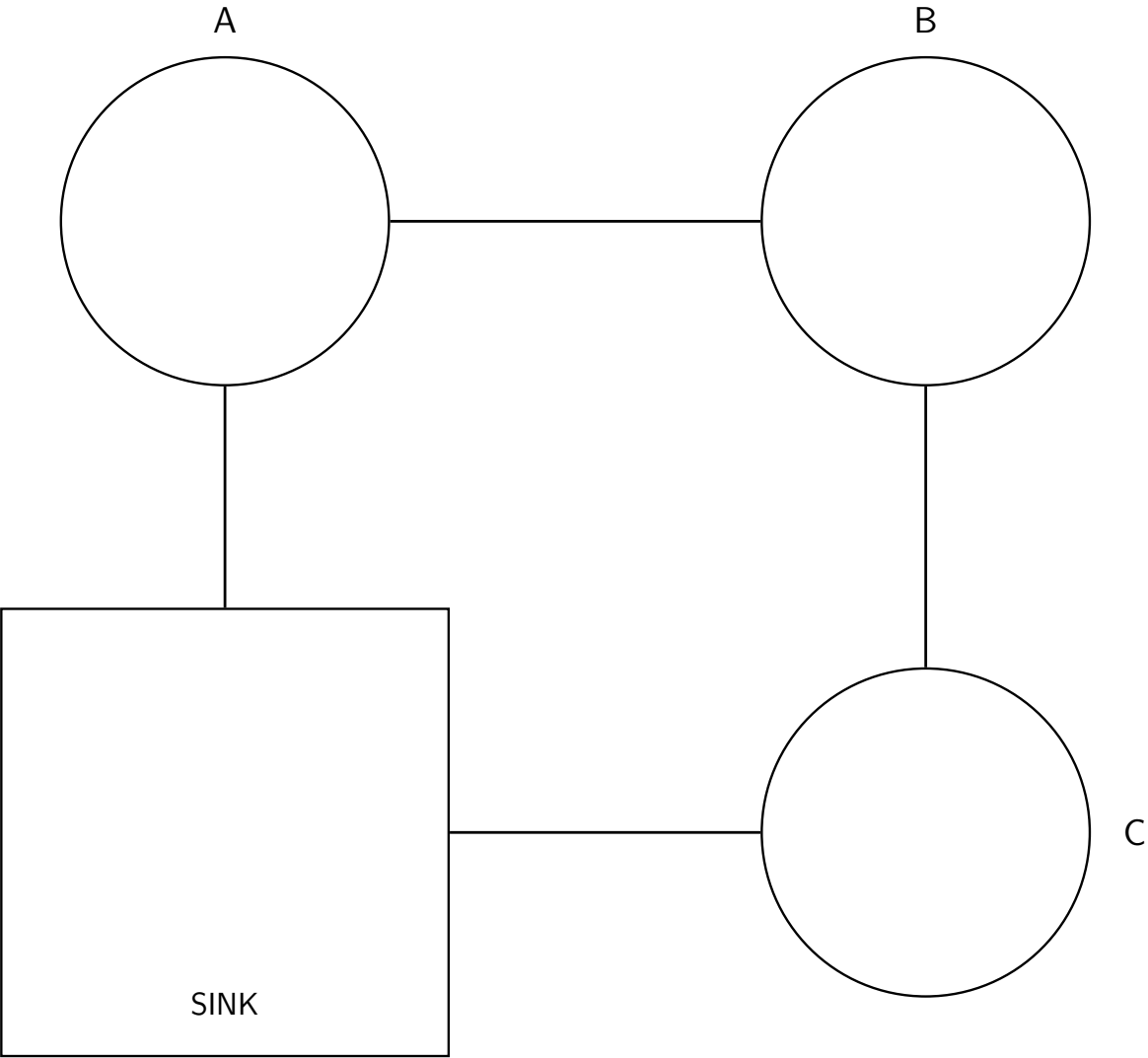
Start A	Start B	Finish A	Finish B

Problem 4: Finish each start. Find another order wherever a choice is possible, and compare the final piles and the number of times each circle shares.



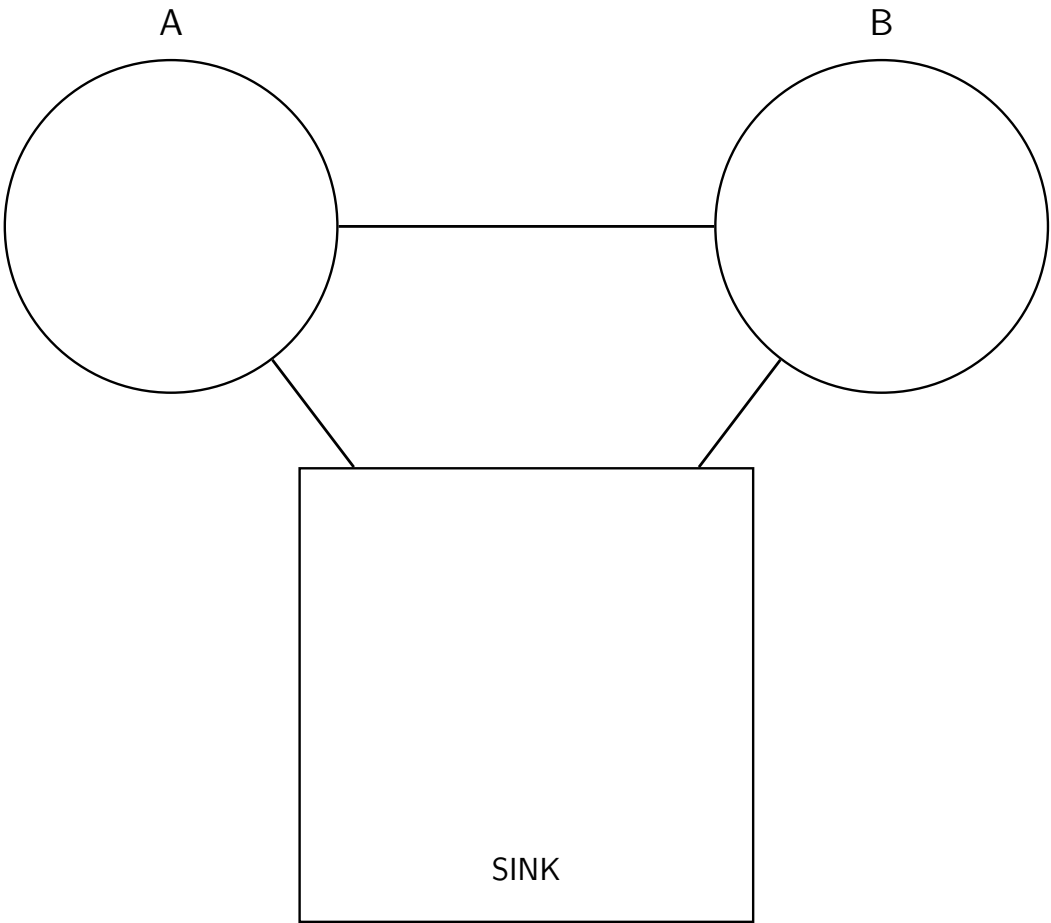
Start A	Start B	Start C	End A	End B	End C	A shares	B shares	C shares
3	0	3						
3	0	3						
0	6	0						
0	6	0						
2	2	2						
2	2	2						
4	1	2						
4	1	2						

Problem 5: Start empty for each trial. Add the two batches together and finish, or finish one batch before adding the other. Try both orders of the batches. Can these choices change the final piles?



Add at A	Add at B	Finish A	Finish B	Finish C
2	4			
2	4			
2	4			

Problem 6: Begin with empty circles. Add one chip to A and finish; keep repeating until you have added 12 chips. Begin again, adding to B instead. After at least one addition, which pairs of final piles can each rule reach?



Chips added	Add to A: final A	Add to A: final B	Add to B: final A	Add to B: final B
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				